

Auxiliary corpus

MIRAGE-AppAct-2024
6.97 GB zip

stream + range reads
2,245 JSON members

reduce 32 shards
→ 90,610 flows · 20 apps

Self-supervised pretraining (8,000 steps)

Masked packet modelling
reconstruct size bin + log-gap

Contrastive NT-Xent
two augmented views attract

Domain-adversarial
aux vs competition, 5-pkt views

Random truncation
5/8/12/20 packets

PacketFormer encoder (frozen for probes)

5 tokens + [CLS]
d=96 · 4 heads · 3 layers

Frozen-encoder linear probes → F12

P1 app family
20-way softmax

P2 media modality
video-likeness

P3 bitrate/pacing
ridge + residuals

P4 padding quantum
(degenerate $\rightarrow$ 0 cols)

P5 embedding PCA
16 components